

17/07/2025

BUJAGALI HYDROPOWER PLANT. UNIT 2 DOWNSTREAM LIFTING BEAM INSPECTION AND LIFTING REPORT.



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1.0 INTRODUCTION.

Diving operations were carried out at unit 2 downstream gate 2 tail race of Bujagali HPP to recover and reposition a lifting beam that was stuck and unbalanced in the tail race channel. The diving team successfully used chain block and lifting bag/ balloon to gradually align and lift the beam.

2.0 GENERAL INFORMATION.

2.1 CLIENTS DETAILS.

Client Company Name	Bujagali Energy Limited
Site Location	Bujagali Hydro Power Station, Jinja
Contact Person	Yusuf, Albert, Moses

2.2 CONTRACTORS DETAILS.

Contractors Company Name	Sun link Engineers and Constructors
Contractors Address	Main Street 4 Jinja
Contact Person	Diving Engineer Mustafa Lwanga

2.3 SITE DETAILS.

Scope	Inspection and alignment of downstream lifting beam	Water Depth (m)	11
Issued Unit(s)	Unit 2 Downstream	Diving Date	19/07/2025
Visibility	Medium	Water Current	High at the surface
Altitude	0° 29'54.00" N, 33° 08.00" E (latitude:0.498325; longitude 33.137500)	Temperature	22 °C

2.4 DIVE TEAM.

Dive Supervisor	Ahmed Fankupi
Divers, Tender & Tech	Abdallah Ahmed, Glyn Mtana, Raza Akram, Ngobi Koowa Joseph

3.0 SCOPE:

The purpose of the operation was to recover and align a stuck lifting beam located in the tail race section. The beam had shifted and was resting at an angle, obstructing normal flow of lifting up/ down of the stop log and posing operational hazards. This involved;

- i. Underwater inspection of the lifting beam.
- ii. Securing lifting points and rigging.
- iii. Controlled lifting and alignment using chain blocks and lifting bag.

4.0 PHOTOGRAPHIC SECTION.



Table 1: Showing how the lifting beam edges were in contact with the guide rails.

5.0 METHODOLOGY.

- i. Pre-dive safety briefing and risk assessment.
- ii. Diver to inspect and clear debris around the lifting beam if any.
- iii. Install chain blocks and slings to main lifting points.
- iv. Attach and position lifting bags for buoyancy assistance.
- v. Gradually apply lift using chain blocks, adjusting lifting bags to balance.
- vi. Align beam horizontally and secure with surface crane/hoist.
- vii. Final checks and removal of rigging.

6.0 DESCRIPTION OF DIVING OPERATIONS.

The diving team mobilized on 19th July 2025 at 11:30. After a safety briefing and equipment checks, the divers entered the water to locate and inspect the lifting beam. The beam was confirmed to be stuck and not properly balanced on the tail race/ guide rails at around 11 meters. The upper side of the lifting beam edge was stuck at 11.6 meters deep, leaving a gap of about 5 centimeters towards the guide rail, while the lower side was stuck at 11.2 meters deep, leaving no gap between the lifting beam edge and the guide rails. This resulted in an inclination of approximately 0.4 meters. Divers installed slings and chain block of 2 tons on the available lifting points, and inflatable lifting bag/ balloon of 1 ton was positioned strategically. A controlled lift was executed in stages to realign the beam before final lifting up.

7.0 EQUIPMENT USED.

1. Surface-supplied diving equipment.
2. Underwater communication system.
3. Chain blocks 3.0 tons.
4. Inflatable lifting bags 1 ton
5. Slings & shackles etc.

8.0 CHALLENGES / AREAS OF CONCERN.

- Initial imbalance of the beam made rigging difficult, there was no up/down movement.
- Surface water current and turbulence due to open spillways required careful diver positioning.

9.0 RESULTS & FINDINGS.

The lifting beam was successfully recovered and aligned without damage. No injuries or incidents were reported. Minor scratches were noted on the guide railings on both walls/ sides.

10.0 SAFETY MEASURES DURING OPERATION.

- i. Continuous surface-supplied air monitoring and standby diver on deck.
- ii. Use of underwater communication and diver umbilical.
- iii. Gradual lifting to prevent sudden shifts.

11.0 RECOMMENDATIONS.

1. Conduct underwater inspection on the guide rails to check its alignment especially where the lifting beam was stuck.
2. Schedule such operations during low-flow periods/ no surface currents.